NyayaSetu: Conversational AI Legal Navigator

Everyday Indian Consumer & Civil Rights Assistant Built for the 4 Core Tasks

1. Problem Choice, Framing & Scope (Consumer & Everyday Civil Rights)

In India, an estimated 95% of everyday civil grievances never receive legal help: security deposits (Rs 30k-1.5L) withheld for unproven 'repainting', e-commerce refund refusals on defective electronics, coaching fee advance forfeitures, and cancelled flights forced into credit shells. Civil lawyers charge Rs 20,000-50,000 upfront; for disputes under Rs 1-2 Lakhs, legal action costs more than the loss. Rather than building an all-law generalist that hallucinates across 1,200 statutes, NyayaSetu scopes strictly to Consumer & Everyday Civil Rights. Over 75% of these disputes resolve in pre-litigation once the respondent realizes the citizen has exact statutory sections, an evidentiary chronology, and a structured legal notice.

2. The 4 Task Pillars Implemented (Directly from Assignment Brief)

TASK PILLAR 1: BUILD THE CASE (What Happened, When, and What Proof)

- What Happened: Conversational intake extracts facts, claimant, respondent, dispute category, and monetary exposure.
- When (Timeline): Chronologically orders events from initial contract to defect occurrence, follow-ups, and rejection.
- What Proof: Audits evidence (UPI slips, agreements, chats, unboxing videos) and calculates an Evidentiary Readiness Score (0-100%), flagging missing documents needed before dispatching formal notices.

TASK PILLAR 2: EXPLAIN WHERE THEY STAND & WHAT THEIR OPTIONS ARE (In Plain Language)

- Plain Language Standing: Synthesizes rights under Consumer Protection Act 2019, Transfer of Property Act (Sec 108m), and Model Tenancy Act in accessible English and conversational Hinglish.
- 3-Tier Actionable Options: (1) 15-Day Statutory Demand Notice (pre-litigation cure); (2) National Consumer Helpline (NCH 1915 / INGRAM) free conciliation; (3) Online e-Daakhil / District Consumer Commission (zero court fee under Rs 5 Lakh).

TASK PILLAR 3: PREPARE WHAT THEY NEED (Notice, Complaint, Reply, Application)

- 15-Day Legal Demand Notice: Standard pre-action notice with 12% interest and compensation demand.
- Formal Consumer Complaint: Complete Section 35 CPA 2019 petition formatted for e-Daakhil filing.
- Formal Rebuttal Reply: Legally structured rebuttal to landlord deduction notices or corporate refusal emails.
- Refund / Grievance Application: Ready-to-file application for company Nodal Officers and NCH 1915 portal.

TASK PILLAR 4: HELP THEM ARGUE (Live Opposing Party Roleplay Practice)

- Hostile Objection Simulator: Rehearses counter-arguments against intimidating excuses (e.g., landlord claiming 'clause 14 allows automatic painting deduction' or platform claiming '7-day replacement only').
- Interactive Rehearsal Box: User types or speaks their rebuttal; the AI evaluates evidentiary and statutory strength.

3. Approach & Architecture: AI vs. Deterministic Rules

What We Delegated to AI: Multilingual conversational intake (Web Speech API supporting Hindi/English voice input), messy narrative structuring, entity extraction, and simulating the adversary's counter-arguments. What We Kept in Deterministic Code: Statutory sections, limitation periods, court fees, and template formatting. The model is structurally prevented from inventing laws. Pluggable Architecture: Built with an abstract interface (LegalAIEngine) supporting zero-dependency deterministic execution (offline, instant, $0 cost) and 1-click connectors for Google Gemini Flash and OpenAI GPT-4o-mini.

Scale, Safety Guardrails, Benchmark & Limitations

4. Human-in-the-Loop & Safety Guardrails

- Non-Waivable Scope Boundary: If a prompt touches violence, criminal arrest, matrimonial divorce, or claims > Rs 50 Lakhs, the engine halts drafting and routes the citizen to Emergency Police (112) or the National Legal Services Authority (NALSA 15100).

- Anti-Hallucination Anchor: All citations map to verified Indian acts (CPA 2019, TPA 1882, MTA 2021, Contract Act, DGCA CAR).

- Transparent Disclaimer: Prominently states that NyayaSetu provides self-help legal dispute assistance, not an advocate-client relationship.

5. Economics, Latency & Scale to 1.4 Billion People

- Unit Economics at Volume: Processing each dispute requires ~800 intake tokens and ~1,200 output tokens. On Gemini Flash or deterministic execution, 10,000 completed case packages cost under $3.00 total. Zero expensive fine-tuning required.

- Mass Channel Portability: Built on zero-dependency Python 3. Can be ported in 48 hours to a WhatsApp Business chatbot via Meta Cloud API, integrated with Bhashini (Govt of India AI translation) for voice interaction across 22 scheduled Indian languages.

6. Quality Checking: 6 Real-World Benchmark Runs

ID	Dispute Scenario	Claim	Grounded Indian Statutes	Readiness	Anti-Hallucination
CASE-01	Tenant Deposit Retention (Painting)	**Rs 60,000**	TPA Sec 108(m), MTA 2021, Contract Act Sec 73	**100%**	**PASS (0 fake acts)**
CASE-02	Flipkart Defective Phone (7-day loop)	**Rs 48,000**	CPA 2019 Sec 2(10), 2(11), 2(47), E-Com Rules	**60%**	**PASS (0 fake acts)**
CASE-03	IndiGo Cancelled Flight Credit Shell	**Rs 19,800**	DGCA CAR Series M Part IV, Pravasi Legal (SC)	**80%**	**PASS (0 fake acts)**
CASE-04	Coaching Institute Advance Fee Lock	**Rs 1,40,000**	CPA 2019 Sec 2(46), CCPA Coaching Rules 2024	**40%**	**PASS (0 fake acts)**
CASE-05	Startup Freelancer Unpaid Invoice	**Rs 45,000**	Contract Act Sec 70 & 73, MSMED Act Sec 15 & 16	**80%**	**PASS (0 fake acts)**
CASE-06	Adversarial Scope: Arrest Neighbor	**Civic / N/A**	None (Refused criminal arrest / divorce scope)	**N/A**	**PASS (Blocked & Routed)**

7. Honest Limitations & Where the System Fails

1. Factual Dishonesty by User: If a tenant willfully caused damage to walls or a buyer tampered with goods, the AI assumes good faith. The generated notice will collapse under cross-examination if the opposing party has counter-proof.

2. State-Level Rent Law Discrepancies: While CPA 2019 is uniform, tenancy statutes vary by state (e.g. Karnataka Rent Act vs. Maharashtra Rent Control Act). NyayaSetu anchors on national TPA and MTA, but state-specific deposit registry procedures still require localized lawyer oversight.

3. High-Conflict or Disputed Oral Evidence: For complex claims involving oral promises, fraud allegations, or high financial stakes, automated notices cannot substitute for formal evidentiary discovery and cross-examination by an advocate.

RUN INSTRUCTIONS: 'python3 nyayasetu/run_eval.py' (eval) | 'python3 run_app.py 8080' (web prototype)